

Newsletter 1.

1) In today's newsletter, we're excited to explore the rise of Artificial Intelligence (AI). This transformative technology has quickly become a part of everyday life, whether in professional, academic, or personal situations, and it's now hard to imagine life without it. We'll dive into what AI is, the types that exist, and how it's impacting the Venture Capital (VC) industry.

2) AI imitates human cognitive abilities by recognizing and sorting information from input data. This intelligence can be based on programmed processes or generated through machine learning. AI originally stems from the 1950s, with the creation of early computer programs that aimed to replicate human intelligence. This innovation led to advancements such as machine learning and eventually deep learning, which enabled computer systems to understand language and speech (natural language processing). In recent years, rapid progress has been made in the field of machine learning, particularly due to the increasing availability of data.

AI relies on large training datasets to build models that recognize patterns and make predictions. Using techniques like supervised learning (learning from labeled data) and unsupervised learning (finding patterns in unlabeled data), AI models continuously adapt and improve their accuracy without needing explicit programming.

3) AI tools have fundamentally shifted from building models from scratch to allowing developers to quickly create AI applications with generative features like audio, image, and text production at scale, by building on existing open scale. This is illustrated in the so-called "AI stack", which includes three foundational layers of AI: infrastructure, data, and model. While this structure may vary by application, it generally follows a fundamental framework to support AI deployment.

The first layer, Infrastructure or also called Foundational models, containing general or parametric knowledge, serve as the computational backbone for AI applications. These large-scale models are pre-trained on extensive datasets, allowing them to recognize patterns and understand language structures, forming a versatile base for AI functionalities. For updated and specific knowledge, they rely on external sources such as databases and search engines beyond their initial training data. These sources work together to enable generative AI by blending the broad insights of foundational models with context-specific, real-time data.

The high demand for AI applications drives the essential need for infrastructure providers to enable scalable and efficient deployment. Cloud services like Amazon Web Services (AWS) and Google Cloud lay the foundation to the AI ecosystem, providing the computational platform and resources, such as processing power, data storage, and scalability, needed for AI applications to operate effectively. Without these infrastructure providers, training and deploying AI models would be challenging due to their high computational demands.

Next, the Data Layer plays a crucial role in organizing, storing, and preparing high-quality data required to train AI models effectively. Hugging Face, an ESCP spin-off startup valued at over \$2 billion, is a leader in providing accessible tools for dataset management, model sharing, and fine-tuning. Its platform allows users to access large datasets, making AI development more efficient and adaptable to specific tasks.

The third layer, the Model (Intelligence) Layer, is the most customer-friendly part of the AI stack as it generates human-like text, analyzes datasets, and translates languages. This layer allows models like OpenAI's GPT-4 to interact naturally with users, supporting creative, analytical, and reasoning tasks through a deep understanding of language and data.

In summary, the AI stack provides a scalable framework for integrating AI into everyday activities, fueling the exponential growth in AI demand and delivering significant benefits across industries.

4) The design of the AI stack has quickly made AI an essential tool in professional and educational activities. This rising demand is driven by AI's key benefits: enhanced decision-making, streamlined automation and efficiency, and highly personalized experiences.

Firstly, AI improves decision-making by providing data-driven insights and reducing human error. For example, football clubs (e.g. Liverpool FC) recently started to use TacticAI (Model Layer), which leverages predictive and generative AI to analyze player positioning and recommend optimized corner kick strategies.

Furthermore, AI improves efficiency and automation in the wholesale and supermarket industries. For instance, supermarket chain Tesco launched an AI-driven dynamic pricing tool in December 2023, using machine learning models trained on historical and current sales data to determine optimal pricing strategies. This enables Tesco to quickly respond to market fluctuations and maintain competitive.

Lastly, AI enables personalization by tailoring experiences to individual preferences. Spotify, for instance, uses algorithms to analyze users' listening habits and favorite genres, employing collaborative filtering and audio analysis through the Model Layer. This approach enables its recommendation engine to curate playlists like Discover Weekly, which align with each user's unique preferences.

5) Ultimately, the rise in demand for generative AI has attracted VC and other strategic investors, aiming to capitalize on the opportunity. To illustrate, in 2024 until September, EY found that a third of all VC invested capital is going to AI startups. However, when looking at the recipients of the investments, it appears that a large majority of capital is for infrastructure models, such as those developed by OpenAI, Anthropic or Mistral AI. This is similar to the development of smartphone applications, where the invested capital went towards the underlying technology before going to app developers. The biggest difference between early smartphone investments and today's AI investments lies in who the primary investors are. Currently, big tech companies have outspent traditional Silicon Valley investors on these startups. Microsoft, Google, and Amazon alone are responsible for two-thirds of the \$27 billion raised by AI companies in 2023.

This shift is primarily due to two main reasons: first, VCs slowed down fund deployment to adjust to higher interest rates and falling valuations of their portfolio companies. VC investment in tech reached record levels in 2021, as investors took advantage of the ultra-low interest rates that enabled them to raise large funds and quickly deploy them, especially in industries affected by Covid-19. Second, developing and training generative AI tools is highly resource-intensive, demanding substantial computing power. As a result, startups have leaned towards partnerships with Big Tech companies that offer cloud infrastructure and access to advanced chips, essential for AI development.

Nonetheless, there have been notable VC investments in the space. While soaring demand has quickly inflated valuations, making it tougher for VCs to invest on early-stage, many have focused on later-stage investments. For instance, Paris-based Mistral has raised around \$500 million from investors like Andreessen Horowitz, General Catalyst, and also Nvidia within just a year of its founding.

As VCs and Big Tech companies rush to invest in AI startups, there is growing concern over the sustainability of these returns. These hype-driven investments may lead to unsustainable valuations and returns if these companies cannot deliver genuine, scalable AI solutions. Therefore, it is more important than ever for investors to critically assess whether AI claims are integral to a startup's core operations or merely an 'AI-washing' strategy to capitalize on the current trend.

6) In summary, AI's rapid adoption, from applications like chatbots to advancements in research, is driving transformative benefits across industries. As investments continue to surge, it's clear that AI's momentum shows no signs of slowing down. Yet, rapid growth also raises questions about the sustainability of these returns. We look forward to exploring this in future newsletters!